

OLS:

Ordinary Least Squares (OLS) is a fundamental statistical technique used to estimate the relationship between one or more independent variables (predictors) and a dependent variable (outcome)

Shapiro-Wilk:

A statistical test used to check if a data sample has come from a normally distributed population or not.

- Null Hypothesis(H_0) - That the population is normally distributed
- Alternative Hypothesis (H_1) - The population is not normally distributed.
- P-Value Interpretation: If the p-value is less than significance level (commonly $\alpha = 0.05$), then rejecting the null hypothesis and conclude the data is not normal. If it is greater, fail to reject normality.

Levene's test:

It is an inferential statistic used to assess the equality of variances for a variable calculated for two or more groups. This test is used because some common statistical procedures assume that variances of the populations from which different samples are drawn are equal.

Mann-Whitney U-test:

It also known as the Wilcoxon rank-sum test is a non-parametric hypothesis test used to assess the difference between two independent samples of continuous data. It compares the distributions of two independent groups to check whether one group tends to have higher or lower values than the other.

One-Way ANOVA:

It is a parametric statistical method used to determine whether there is a significant difference among the means of three or more groups by testing the null hypothesis that all group means are equal.

Q-Q plot:

The Quantile-Quantile (Q-Q) plot is a graphical method used to determine whether a dataset follows a specific probability distribution or whether two datasets come from the same population.

Variance Inflation Factor (VIF):

It measures how much the variance of an estimated regression coefficient increases because of collinearity (correlation) among independent variables in a model